

```
In [1]: import numpy as np
from ansys.mapdl.core import launch_mapdl
#exec_loc = "/Program Files (x86)/ANSYS Inc/v222/ansys/bin/winx64/ansys222.exe" #change to your executable location
mapdl = launch_mapdl()
```

```
In [2]: #!!!!!!!!!!!!!!
#! PARAMETERS
#!!!!!!!!!!!!!!!
R=12.6051E-3
H=60E-3
NV=40
NC=37
D_SHELIX=0.97804E-3
AA=0.19453E-3
BB=0.19376E-3
LOOP_NO = 4
PITCH=H/LOOP_NO
PI = np.pi
DELTA_TH = 2*PI/NV
DELTA_H = H/(NC-1)
TH_SHELIX = D_SHELIX/R
NOBUCK = 10
```

```
In [3]: mapdl.prep7()
```

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Out[3]: *** MAPDL - ENGINEERING ANALYSIS SYSTEM  RELEASE 2022 R2          22.2      ***
        Ansys Mechanical Enterprise Academic Student
        00000000 VERSION=WINDOWS x64   18:45:56  JUN 27, 2023 CP=      0.047
```

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In [4]: #!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
#! CROSS SECTION OF THE BIG HELIX
#! AT THE END SECTION IS SAVED AND
#! UNNECESSARY DATA IS DELETED
#!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
mapdl.et(1,'PLANE82')
mapdl.clocal(11,1,"", "", "", "", "", 2)
mapdl.csys(11)
RR = 1.2269E-3
COUNT = 1
for TH in range(0,181,15):
    mapdl.k(COUNT,RR,TH,0)
    COUNT = COUNT + 1

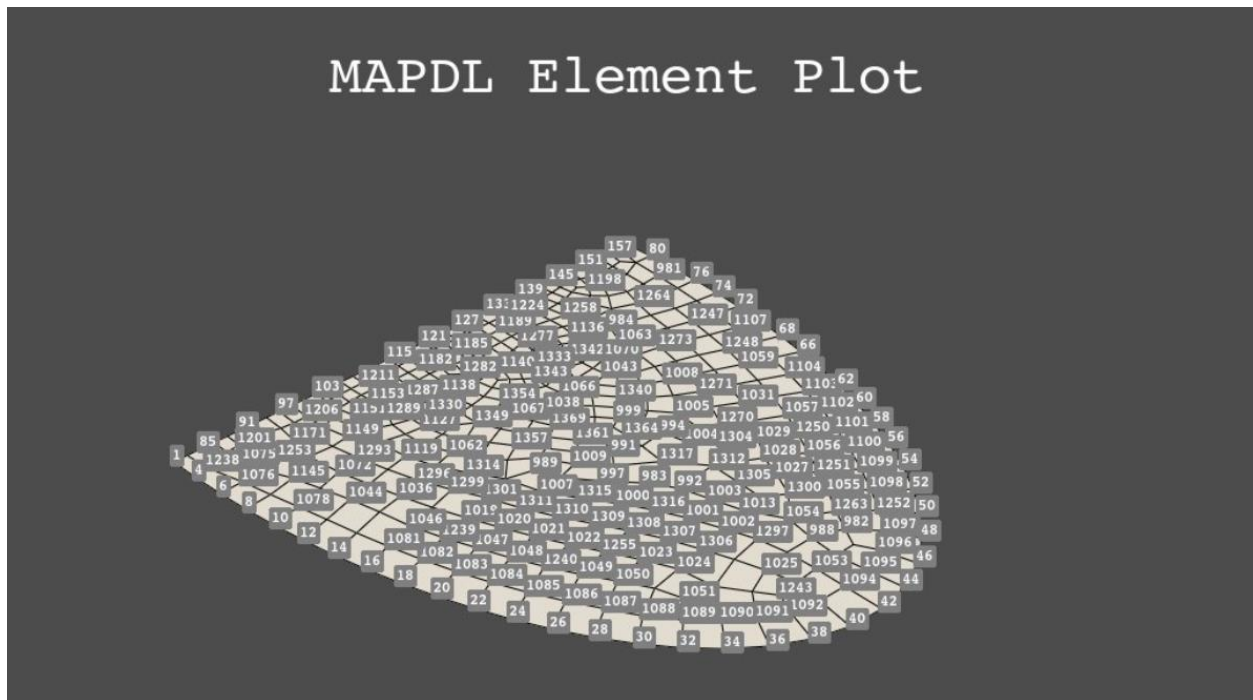
mapdl.flst(3,13,3)
SPLINE_COUNT = 1
for I in range(1,14):
    mapdl.fitem(3,SPLINE_COUNT)
    SPLINE_COUNT = SPLINE_COUNT + 1

mapdl.bsplin("", 'P51X')
mapdl.csys(0)
mapdl.l(1,13)
mapdl.lsize("All", "", "", 40, "", 1, "", "", 1)
mapdl.al(1,2)
mapdl.amesh("All")
mapdl.esel("All")

mapdl.eplot("All") #plot element cross section of big helix

mapdl.secwrite("big_helix")
mapdl.aclear("All")
mapdl.adele("All", "", "", 1)
```

Showing the ellipsoid cross section:



```
In [5]: #!!!!!!!!!!!!!!!!!!!!!!!  
        #! MAIN CODE STARTS  
        #!!!!!!!!!!!!!!!!!!!!!!!  
  
        mapdl.et(1,'BEAM188') #ELEMENT TYPE 3D 2-node beam element  
        mapdl.mptemp("","","","","","")  
        mapdl.mptemp(1,0)  
        mapdl.mpdata("EX",1,"",91900000000)  
        mapdl.mpdata("PRXY",1,"",0.17) #Define material properties  
        mapdl.sectype(1,"BEAM","RECT","",0) #Define rectangular cross section for beam elements  
        mapdl.secoffset("CENT")  
        mapdl.secddata(AA,BB,0,0,0,0,0,0,0,0,0) #Defines size of cross section  
  
        mapdl.sectype(2,"BEAM","MESH") #ELLIPSOIDAL Cross SECTION  
        mapdl.secoffset("ORIG","","")  
        mapdl.secread("big_helix","SECT","MESH")
```

```

In [6]: #!!!!!!!!!!!!!!!!!!!!!!
#!/ VERTICAL BEAMS
#!!!!!!!!!!!!!!!!!!!!!!
COUNT=1
TH=0
for II in range(1,NV+1):
    mapdl.k(COUNT,R*np.cos(TH),R*np.sin(TH),0)
    TH = TH + DELTA_TH
    COUNT = COUNT + 1

TH=0
for II in range(1,NV+1):
    mapdl.k(COUNT,R*np.cos(TH),R*np.sin(TH),H)
    TH = TH + DELTA_TH
    COUNT = COUNT + 1

```

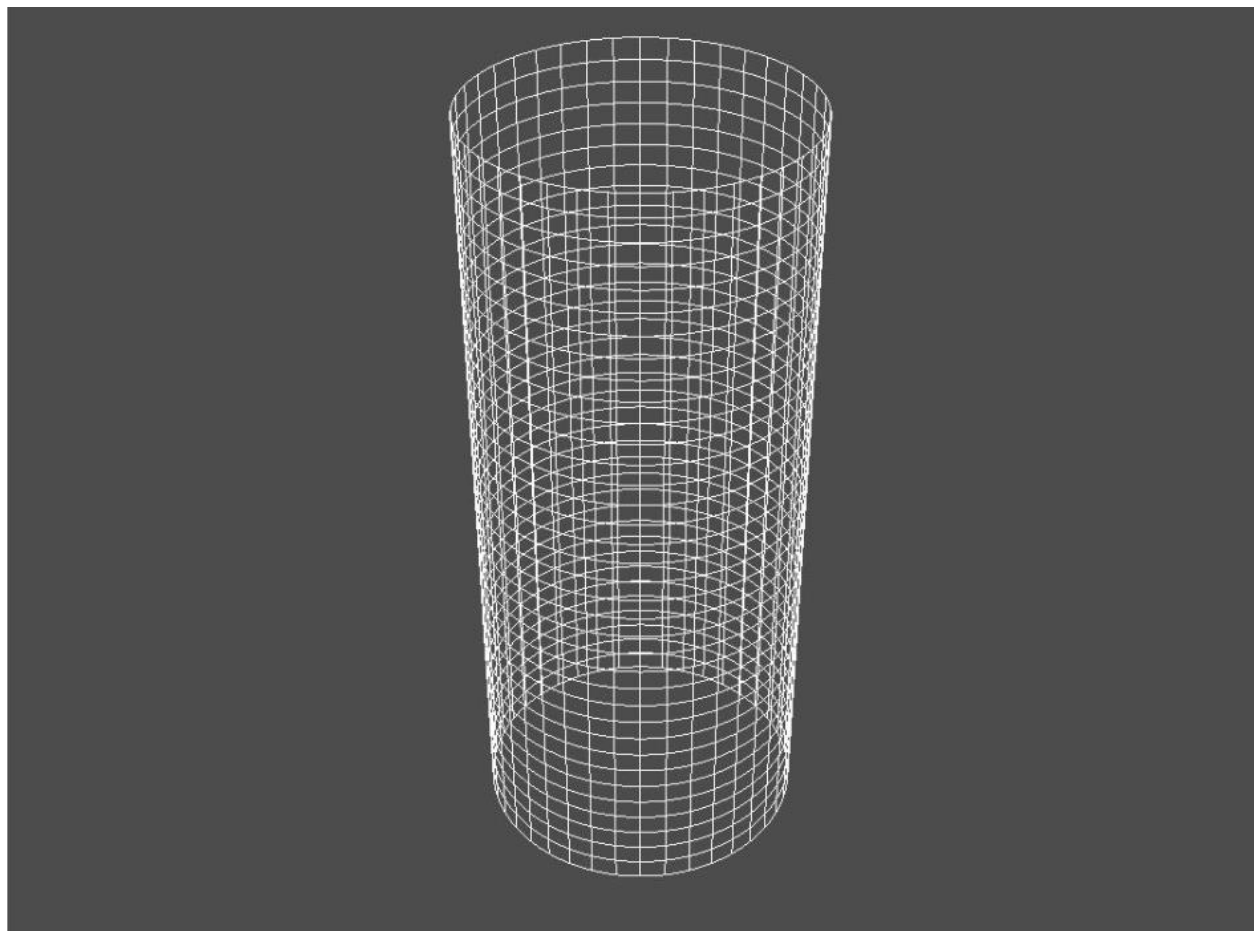
```

In [7]: #!!!!!!!!!!!!!!!!!!!!!!
#!/ CIRCUMFERENTIAL BEAMS
#!!!!!!!!!!!!!!!!!!!!!!
COUNT_LINE=1
for II in range(1,NV+1):
    mapdl.l(COUNT_LINE,COUNT_LINE+NV)
    COUNT_LINE = COUNT_LINE + 1

HH=0
for II in range(1,NC+1):
    mapdl.k(COUNT,0,0,HH)
    mapdl.circle(COUNT,R)
    COUNT = COUNT + 5
    COUNT_LINE = COUNT_LINE + 4
    HH = HH + DELTA_H

mapdl.lplot("All")

```



```

In [8]: #!!!!!!!!!!!!!!!!!!!!!!
        #!! SMALL HELICES 1st direction
        #!!!!!!!!!!!!!!!!!!!!!!
        TH0 = TH_SHELIX/2
        TH1 = -TH_SHELIX/2
        HH=0

        for JJ in range(1,int(NV/2)+1):
            SPLINE_INIT = COUNT
            HH=0
            for II in range(1,NC+1):
                mapdl.k(COUNT,R*np.cos(TH0),R*np.sin(TH0),HH)
                COUNT = COUNT + 1
                TH0 = TH0 + DELTA_TH
                HH = HH + DELTA_H
            mapdl.flst(3,NC,3)
            for II in range(1,NC+1):
                mapdl.fitem(3,SPLINE_INIT)
                SPLINE_INIT = SPLINE_INIT + 1
            mapdl.bsplin("", 'P51X')

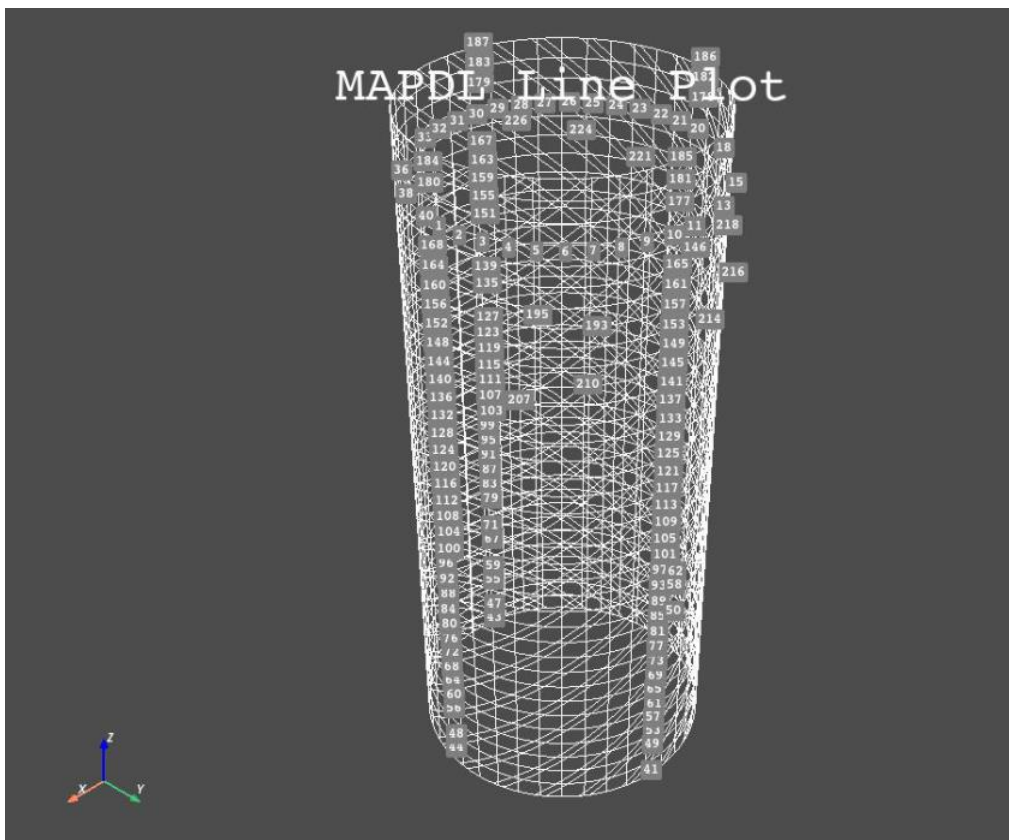
            COUNT_LINE = COUNT_LINE + 1
            SPLINE_INIT = COUNT
            HH=0
            for II in range(1,NC+1):
                mapdl.k(COUNT,R*np.cos(TH1),R*np.sin(TH1),HH)
                COUNT = COUNT + 1
                TH1 = TH1 + DELTA_TH
                HH = HH + DELTA_H
            mapdl.flst(3,NC,3)

            for II in range(1,NC+1):
                mapdl.fitem(3,SPLINE_INIT)
                SPLINE_INIT = SPLINE_INIT + 1
            mapdl.bsplin("", 'P51X')

            COUNT_LINE = COUNT_LINE + 1
            TH0 = TH_SHELIX/2+(2*DELTA_TH)*JJ
            TH1 = -TH_SHELIX/2+(2*DELTA_TH)*JJ

        mapdl.lplot("All")

```



```

In [9]: # #!!!!!!!!!!!!!!!!!!!!!!
      #! SMALL HELICES 2nd direction
      # #!!!!!!!!!!!!!!!!!!!!!!
      TH0 = TH_SHELIX/2
      TH1 = -TH_SHELIX/2
      HH=0

      for JJ in range(1,int(NV/2)+1):
          SPLINE_INIT = COUNT
          HH=0
          for II in range(1,NC+1):
              mapdl.k(COUNT,R*np.cos(TH0),R*np.sin(TH0),HH)
              COUNT = COUNT + 1
              TH0 = TH0 - DELTA_TH #sign switches
              HH = HH + DELTA_H
          mapdl.flst(3,NC,3)
          for II in range(1,NC+1):
              mapdl.fitem(3,SPLINE_INIT)
              SPLINE_INIT = SPLINE_INIT + 1
          mapdl.bsplin("", 'P51X')

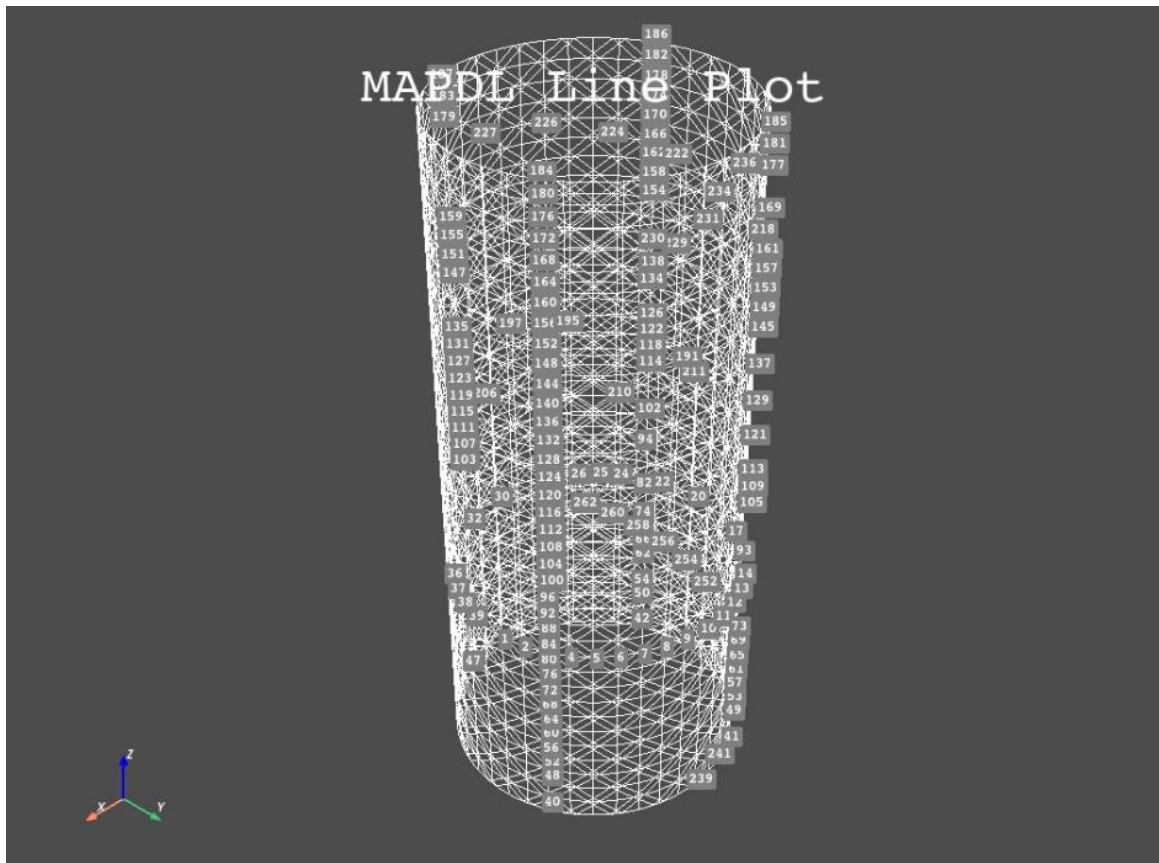
          COUNT_LINE = COUNT_LINE + 1
          SPLINE_INIT = COUNT
          HH=0
          for II in range(1,NC+1):
              mapdl.k(COUNT,R*np.cos(TH1),R*np.sin(TH1),HH)
              COUNT = COUNT + 1
              TH1 = TH1 - DELTA_TH #sign switches
              HH = HH + DELTA_H
          mapdl.flst(3,NC,3)

          for II in range(1,NC+1):
              mapdl.fitem(3,SPLINE_INIT)
              SPLINE_INIT = SPLINE_INIT + 1
          mapdl.bsplin("", 'P51X')

          COUNT_LINE = COUNT_LINE + 1
          TH0 = TH_SHELIX/2+(2*DELTA_TH)*JJ
          TH1 = -TH_SHELIX/2+(2*DELTA_TH)*JJ

      mapdl.lplot("All")

```



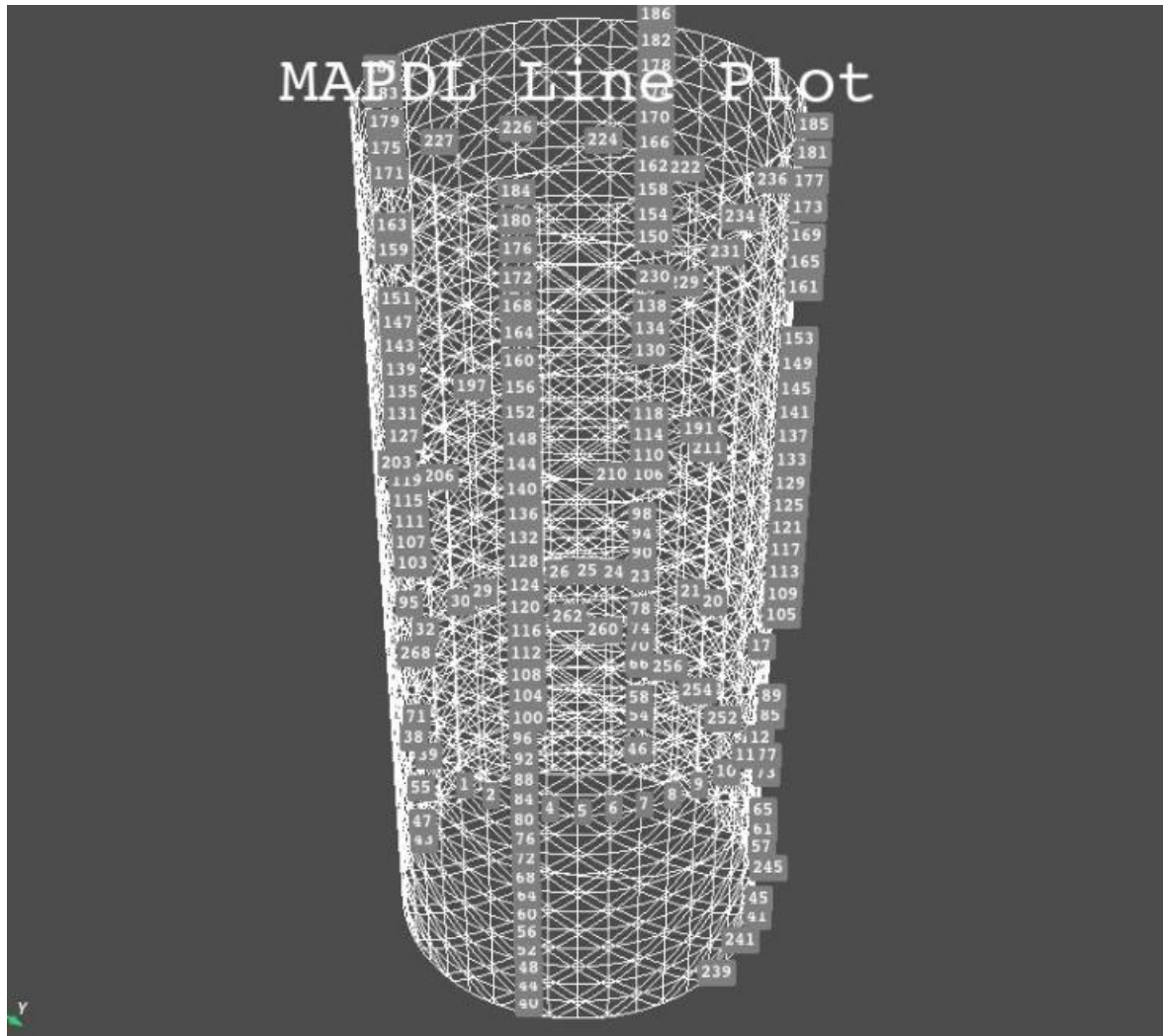
Large Helix with ellipsoid cross section:

```
In [10]: #!!!!!!!!!!!!!!!!!!!!!!
#!! Large HELICE
#!!!!!!!!!!!!!!!!!!!!!!

NUM_KP = mapdl.get('NUM_KP','KP',0,'NUM','MAXD')
COUNT = NUM_KP + 1
HH = 0
TH0 = 0
DELTA_H_BIGHELIX = PITCH/NV
SPLINE_INIT = COUNT

for II in range(1,(NV)*LOOP_NO+2):
    mapdl.k(COUNT,R*np.cos(TH0),R*np.sin(TH0),HH)
    TH0 = TH0 + DELTA_TH
    HH = HH + DELTA_H_BIGHELIX
    COUNT = COUNT + 1
mapdl.flst(3,(NV)*LOOP_NO+1,3)

for II in range(1,(NV)*LOOP_NO+2):
    mapdl.fitem(3,SPLINE_INIT)
    SPLINE_INIT = SPLINE_INIT + 1
mapdl.bsplin('','P51X')
mapdl.lplot("All")
```




```
In [11]: mapdl.isel("All")
mapdl.btol("1e-10")
mapdl.lovlap("All") #Merge all overlapping beams in one structure
print("Beams merged")
```

Beams merged

```
In [12]: mapdl.isel("All")
mapdl.lmesh("All") #Meshing & Plot of Mesh
mapdl.eplot(show_edges=True, smooth_shading=True, show_node_numbering=False)
```

MAPDL Element Plot

